

Dave's Daily

AI • Bayesian optimization US equities

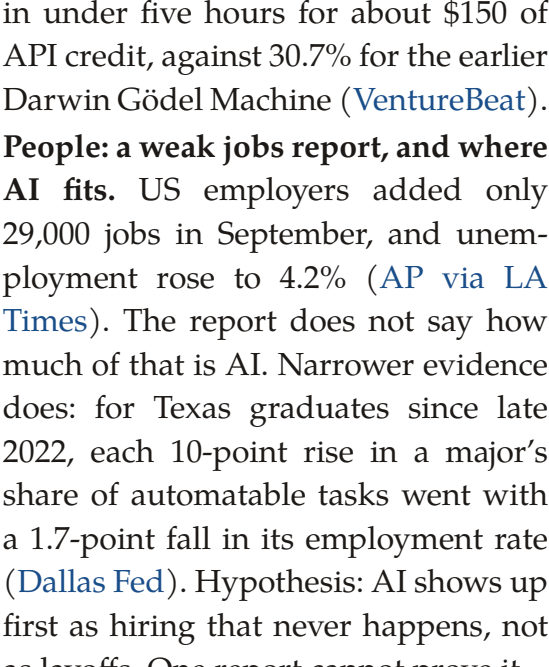
tldr

1. Anyone can now download a model that builds working exploits. Anthropic found open-weight GLM-5.3 nearly matches its restricted Mythos Preview, and its safeguards fall 64–100% of the time.
2. Amazon is reportedly moving about \$8 billion of Nvidia chips into an investor-funded vehicle and renting them back. The bet on how long a GPU stays valuable moves with them.
3. Among simulated language-model traders, rewording one layoff announcement raised one model family's share of orders from 17.5% to 57.4%.

AI

The exploit gap closed in the open. On 29 September, Anthropic's Frontier Red Team tested GLM-5.3, an open-weight model from China's Zhipu AI (Z.ai) ([Anthropic](#)). On ExploitBench, which asks for working attacks on known flaws in Chrome's V8 engine, GLM-5.3 succeeded in 50 of 410 attempts. Claude Mythos Preview, which Anthropic releases only to vetted defenders, succeeded in 56. A red-team cover story got past GLM-5.3's refusals 64% of the time, pre-writing its reasoning 92%, and editing refusals out of the weights 100%. That edit cost Anthropic about \$4,400 of compute. A smaller sibling chained two public Chrome flaws into a working attack for \$20.40. Anthropic sells closed models, so it has a stake here. Still, NIST's CAISI independently rates GLM-5.3 the most cyber-capable open-weight model yet, about four months behind the US frontier.

Google holds its newest model back. Google announced Gemini 4 Argon on 30 September, but only vetted cyber defenders get it for now, through a program called Fairwind ([Google](#)). Reuters reported that the launch came a day after lab chiefs signed a voluntary accord at the White House to police their own systems ([Reuters via Dawn](#)). OpenAI's pause on training, evaluation, and tool use for its most capable models is still in force ([The Verge](#)).



Cheaper self-improvement. In SIFT, from MIT and Sakana AI, a coding agent rewrites itself, and a language-model judge compares pairs of versions. Only the top-ranked ones get a full benchmark run ([arXiv:2609.19526](#)). On the Polyglot benchmark, one run reached 35.1% in under five hours for about \$150 of API credit, against 30.7% for the earlier Darwin Gödel Machine ([VentureBeat](#)).

People: a weak jobs report, and where AI fits. US employers added only 29,000 jobs in September, and unemployment rose to 4.2% ([AP via LA Times](#)). The report does not say how much of that is AI. Narrower evidence does: for Texas graduates since late 2022, each 10-point rise in a major's share of automatable tasks went with a 1.7-point fall in its employment rate ([Dallas Fed](#)). Hypothesis: AI shows up first as hiring that never happens, not as layoffs. One report cannot prove it.

Markets

Renting back the GPUs. The Financial Times reported Friday that Amazon is offering investors a vehicle holding about \$8 billion of Nvidia Grace Blackwell chips ([FT via Financial Post; Bloomberg Law](#)). The vehicle would borrow to buy the chips and lease them back to Amazon. Investors expect an investment-grade rating resting on Amazon's double-A credit, which would let insurers and pension funds in. Amazon would keep no equity. Its July bond sale met weak demand for long-dated debt, and the 10-year Treasury yield is at a 24-year high ([Reuters](#)). The real question is depreciation. Nvidia's Vera Rubin will overtake Grace Blackwell, while Amazon argues each chip generation should last at least five years. Whoever holds the hardware holds that bet.

Whose model is in the order book. Victoria Li and Arka Bandyopadhyay ran 15 language-model traders, five each from Llama, Qwen, and Mistral, in a simulated stock market ([arXiv:2610.01897](#)). Changing only the wording of the news changed which family traded. For a layoff announcement, explanatory wording raised Mistral's share of submitted orders from 17.5% to 57.4%, though it was always one-third of the agents. When a single family filled the whole market, it traded all one way: all-Mistral markets only bought, and all-Qwen markets only sold. The models were small and deep background liquidity pinned prices. The point for algorithmic desks: having diverse models installed does not mean diverse orders reach the book. No new AI result in high-frequency trading appeared in a high-quality source over the past day.

Key numbers (this edition)	
Item	Figure
ExploitBench, GLM-5.3 / Mythos	50 / 56 of 410
Cost to strip GLM-5.3 refusals	~\$4,400
Amazon chips in the vehicle	~\$8 billion
Mistral share, before / after	17.5% / 57.4%
September payroll gain	29,000

Bayesian optimization

Scaling. When a language-model agent steers Bayesian optimization, it stops being reliable as the number of variables grows, report Zhixuan Gao and colleagues at Nanjing University ([arXiv:2609.34281](#)). Their agent, HERA, picks the method rather than the points: which high-dimensional strategy and modeling assumption to try, and for how long. A numerical engine picks the points. On eight real tasks of 60 to 256 dimensions (rover paths, car design, a simulated ant robot, molecules), it had the best mean final result on most.

Data efficiency. PRISMS, posted 1 October, replaces absolute scores with pairwise rankings: "is A better than B?" ([arXiv:2610.01827](#)). Rankers range from rules of thumb to agent workflows to scientists, and a Fisher-information rule decides when to escalate to a costlier one. In drug-library screening, it reached 50% recall of the top 10 in about 42% fewer rounds than regression-based active learning. When generating new designs, it beat a Bayesian optimization baseline's hypervolume by about 18.8%. Separately, putting a cheap physics proxy in the Gaussian-process prior mean raised cell-tower coverage by up to about 15 points at the same budget ([arXiv:2610.00974](#)).

Connections to AI. PRISMS and SIFT (in the AI section) make the same move: when scores are expensive, ask a cheaper judge which of two candidates is better, and rank from the comparisons. HERA moves the language model up a level, from choosing points to choosing the method.